Explainable Multimodal Longitudinal Medical Imaging for Disease Progression

Technical Specification V1

1. Project Objective

The objective of this project is to develop a technical research prototype capable of analyzing the evolution of a patient across multiple medical visits by jointly using longitudinal medical imaging and complementary clinical information.

The system should not only estimate disease progression risk, but also make the evolution of that estimate inspectable. In particular, it should provide information about:

- how the predicted progression risk evolves as new visits become available;
- which visits or transitions between visits contribute most to a change in prediction;
- which anatomical regions of the medical images are relevant;
- which clinical variables influence the model;
- how imaging and clinical information jointly contribute to the final decision.

The initial use case will focus on a single disease-progression task, with Mild Cognitive Impairment (MCI) progression toward Alzheimer's disease as the first candidate scenario.

2. Context

Many machine-learning systems applied to medical imaging analyze an examination independently. However, progressive diseases are observed through repeated visits, and clinically relevant information may lie in how the patient changes over time rather than in one isolated scan.

A longitudinal patient record may contain repeated MRI examinations, cognitive assessments, demographic variables, biological or clinical measurements, and non-uniform intervals between visits. The project therefore represents the patient as a longitudinal multimodal sequence.

3. Technical Problem and Mathematical Formulation

The technical problem is to learn a predictive representation from heterogeneous patient information distributed across both modalities and time, while preserving enough structure to analyze how the prediction was obtained.

3.1 Patient-level representation

For patient i, let the longitudinal record contain T_i visits:

$$\mathcal{P}_i = \{(X_{i1}, C_{i1}, t_{i1}), \ldots, (X_{iT_i}, C_{iT_i}, t_{iT_i})\}$$

where X_it is the medical image acquired at visit t, C_it is the vector of clinical variables available at that visit, t_it is the acquisition time, and T_i is the number of visits available for patient i.

For the initial binary progression task, the target is defined as:

$$Y_i \in \{0, 1\}$$

The exact prediction horizon and cohort definition must be fixed before model training and must remain identical across all baselines.

3.2 Modality-specific representations

Each modality is first mapped to a compact representation:

$$z_{it}^{\text{img}} = f_{\theta_I}(X_{it})$$

$$z_{it}^{\text{clin}} = g_{\theta_C}(C_{it})$$

The imaging encoder f_thetaI and clinical encoder g_thetaC may use different architectures because the underlying data structures are different.

3.3 Visit-level multimodal fusion

For the V1 architecture, imaging and clinical representations are fused at each visit before temporal modeling:

$$z_{it} = \phi_{\theta_F}\left(z_{it}^{\text{img}}, z_{it}^{\text{clin}}\right)$$

This is the initial architectural choice. Alternative designs, such as modality-specific temporal modeling followed by late fusion, can be evaluated later if required.

3.4 Longitudinal representation

The visit representations form an ordered sequence that is processed by a temporal encoder:

$$H_i = g_{\theta_T}(z_{i1}, z_{i2}, \ldots, z_{iT_i})$$

When irregular visit timing is modeled explicitly, the temporal encoder may additionally receive the elapsed interval:

$$\Delta t_{it} = t_{it} - t_{i,\,t-1}$$

3.5 Progression probability and risk trajectory

The system should be able to estimate progression risk not only after the final visit, but after each prefix of the patient history. For visit t:

$$p_{it} = \Pr\left(Y_i = 1 \mid \mathcal{P}_i^{1:t}\right)$$

where P_i^(1:t) contains all visits available from the first visit through visit t. The same trained model is evaluated on progressively longer patient histories; separate models are not required for each visit.

The change in predicted risk between two consecutive visits is defined as:

$$\Delta p_{it} = p_{it} - p_{i,\,t-1}$$

This quantity provides a simple operational definition of when the model's estimated risk changes most strongly. It does not by itself explain why the change occurred; the explainability and change-analysis modules are used for that purpose.

3.6 Initial optimization objective

For a binary progression task, the initial training objective is binary cross-entropy:

$$\mathcal{L}_{\text{pred}} = -\frac{1}{N}\sum_{i=1}^{N}\left[Y_i \log(\hat{p}_i) + (1 - Y_i)\log(1 - \hat{p}_i)\right]$$

Additional temporal, calibration, or explainability regularization terms should only be introduced after they are explicitly defined and justified. They are therefore outside the V1 core objective.

4. Main Expected Result

The final technical result should be an interactive longitudinal patient-analysis prototype. Given the data available for one patient across several visits, the system should provide the following outputs.

Example visible outputs

Progression probability or risk score for the defined task.

Risk trajectory showing how the estimate changes as new visits are added.

MRI evolution across visits and visualization of relevant anatomical regions.

Temporal importance highlighting influential visits or transitions.

Clinical-variable contribution and modality-level contribution.

Model exploration through controlled removal of a visit, modality, or selected clinical information.

5. Intended Technical Product

The intended product is not a medical diagnostic application. It is a research prototype for developing, evaluating, and interpreting longitudinal multimodal disease-progression models.

The visible application should expose five main views:

Patient Overview: patient visits, examination dates, available modalities, clinical measurements, and current prediction.

Progression Timeline: risk evolution, clinical-score evolution, and major transitions between visits.

Imaging Evolution Viewer: side-by-side or longitudinal comparison of medical images and relevant regions.

Explainability Panel: spatial attribution, temporal attribution, clinical-variable contribution, and modality contribution.

Model Exploration Panel: controlled removal or masking of visits, modalities, or selected variables to inspect model behavior.

6. Work to Be Performed

6.1 Phase 1 - Dataset and longitudinal cohort construction

Identify an appropriate longitudinal medical dataset. ADNI is an initial candidate because it contains repeated MRI examinations and longitudinal clinical/cognitive information. Dataset access, cohort definition, and permitted use must be confirmed before implementation.

The first technical task is to construct patient-level sequences from the available visits. Work includes:

- patient selection and cohort definition;
- visit matching and temporal ordering;
- MRI preprocessing;
- clinical-data cleaning and preprocessing;
- normalization and categorical encoding;
- missing-data characterization;
- patient-level train/validation/test splitting to prevent leakage across visits.

7. System Architecture

The V1 system is organized as one end-to-end multimodal longitudinal prediction system composed of modular encoders, a fusion block, a temporal model, a prediction head, and analysis modules. The modules may be developed and validated separately, but the prediction path can later be fine-tuned jointly.

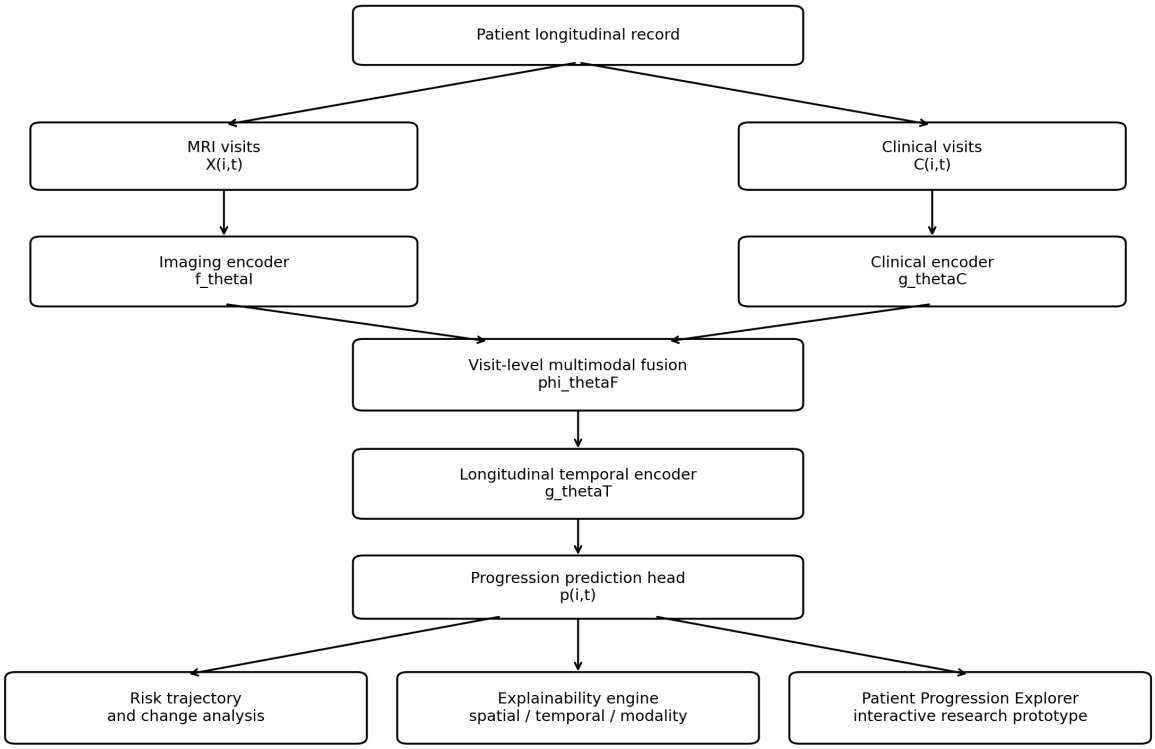


Figure 1. Global architecture of the proposed V1 system.

7.1 Imaging encoder

Purpose: convert each medical image X_it into a compact representation z_img(i,t). The initial baseline should use one reliable 3D imaging encoder. Pretrained or self-supervised alternatives can be evaluated later.

7.2 Clinical encoder

Purpose: transform the clinical vector C_it into a learned representation z_clin(i,t). The first implementation can use normalized numerical variables, encoded categorical variables, and a compact multilayer perceptron.

7.3 Visit-level multimodal fusion

Purpose: combine imaging and clinical representations available at the same visit. The V1 baseline should use a simple and inspectable fusion rule, such as concatenation followed by a projection layer. Gated fusion, cross-attention, or more complex interaction mechanisms are later comparison points rather than V1 requirements.

7.4 Longitudinal temporal encoder

Purpose: model the ordered sequence z_i1, ..., z_iT_i and produce contextualized patient representations. A GRU or temporal Transformer can serve as the initial baseline. Explicit time-interval encoding is an extension if the cohort contains materially irregular visit spacing.

7.5 Prediction head

Purpose: map the longitudinal representation to the probability of the predefined progression event. The prediction head must support prefix-based inference so that p_i1, p_i2, ..., p_iT_i can be used to construct the risk trajectory.

7.6 Explainability engine

The V1 explainability engine is an analysis layer around the trained model rather than a separate predictive model. It should support:

- spatial attribution for imaging inputs;
- temporal contribution of visits or sequence segments;
- clinical-feature attribution;
- modality ablation to compare imaging-only, clinical-only, and multimodal behavior.

Possible post-hoc methods include Integrated Gradients, occlusion analysis, Grad-CAM where architecturally appropriate, SHAP for selected tabular analyses, and controlled ablation tests. The exact method should be chosen based on the final architecture and evaluation goal.

7.7 Change-based analysis

A central extension is to analyze transitions between visits rather than only isolated states. A simple latent difference may be computed as:

$$\Delta z_{it} = z_{it} - z_{i,t-1}$$

This difference is only a representation-space comparison; it must not be interpreted as a clinically meaningful disease-change vector without further validation. Change analysis should combine latent changes, risk changes Delta p_it, and explanation methods to investigate why a specific transition altered the prediction.

8. Baselines

The complete system must be compared with progressively richer baselines so that the contribution of longitudinal and multimodal information can be measured rather than assumed.

Baseline	Imaging	Clinical	Longitudinal
A - Single-visit imaging	Yes	No	No
B - Single-visit multimodal	Yes	Yes	No
C - Longitudinal imaging	Yes	No	Yes
D - Longitudinal clinical	No	Yes	Yes
Proposed V1	Yes	Yes	Yes

9. Evaluation

9.1 Predictive performance

- AUROC and AUPRC;
- F1-score;
- sensitivity and specificity;
- balanced accuracy.

9.2 Calibration

- Brier score;
- calibration curve;

- Expected Calibration Error, if sample size permits stable estimation.

9.3 Temporal contribution

- comparison with latest-visit-only inference;
- removal of previous visits;
- masking selected temporal segments;
- controlled order perturbation as a diagnostic test, not as a clinically meaningful input.

9.4 Multimodal contribution

- imaging-only evaluation;
- clinical-only evaluation;
- multimodal evaluation;
- controlled missing-modality tests when technically supported.

9.5 Explainability evaluation

- fidelity to model behavior;
- stability under small input perturbations;
- sensitivity to feature or visit removal;
- consistency across similar patient visits.

10. Interactive Prototype

The system should expose its results through an interactive research interface implemented, for example, with Streamlit or Gradio. The interface is a visualization and model-inspection layer; it is not a diagnostic medical device.

Proposed workflow

1. Select or load an authorized patient record.
2. Display available visits and modalities.
3. Run longitudinal inference on successive visit prefixes.
4. Display the risk trajectory p_i1 -> ... -> p_iT_i.
5. Inspect MRI evolution and spatial explanations.
6. Inspect temporal and clinical contributions.
7. Compare modality or visit ablations.

11. Technical Deliverables

ID	Deliverable	Expected output
D1	Longitudinal Dataset Builder	Reproducible patient-level sequence construction and preprocessing pipeline.
D2	Baseline Modeling Framework	Single-visit, longitudinal, unimodal, and multimodal baseline implementations.
D3	Multimodal Longitudinal Prediction Model	Working model integrating medical imaging, clinical information, and temporal modeling.
D4	Explainability Engine	Reusable spatial, temporal, clinical-feature, and modality-analysis functions.
D5	Patient Progression Explorer	Interactive research prototype for timeline, risk, imaging, and explanation inspection.
D6	Reproducible Repository	Documented code, configuration files, tests, and reproducibility instructions; restricted datasets are not redistributed.

12. Development Priorities

12.1 MVP

The minimum working system should support longitudinal MRI and longitudinal clinical variables as input and return a progression probability, a visit-by-visit risk trajectory, a basic spatial explanation, and temporal contribution estimates.

12.2 Second version

- improved multimodal fusion;
- modality attribution;
- controlled missing-modality handling;
- change-between-visits analysis.

12.3 Advanced research directions

- cross-modal interaction explanations;
- uncertainty estimation;
- counterfactual trajectories;
- explicit irregular-time modeling;
- continuous-time architectures;
- concept-based explainability.

13. Expected Impact and Utility

The technical value of the project lies in integrating longitudinal medical-image processing, time-series modeling, multimodal fusion, explainable AI, and interactive model inspection into one reproducible system. Rather than

producing only a classification result, the prototype is designed to make the evolution of the model prediction inspectable across a patient trajectory.

The project can serve as a research prototype, a benchmarking framework for longitudinal multimodal models, an explainability experimentation platform, and a technical portfolio demonstrating medical imaging, temporal learning, multimodal AI, and research engineering skills.

14. Core Technical Vision

Core technical vision

Build a system that does not only predict whether a disease is progressing, but allows us to inspect how the patient's multimodal trajectory led the model to that prediction.